

We place triangle ABC in the coordinate plane with $B = (0, 0)$, $C = (17, 0)$. Since $AB = 26$ and $AC = 25$, solving $x_A^2 + y_A^2 = 26^2$ and $(x_A - 17)^2 + y_A^2 = 25^2$ gives $A = (10, 24)$. The orthocenter H is the intersection of altitudes: altitude from A is $x = 10$, altitude from B is perpendicular to AC (slope $-24/7$ so altitude slope $7/24$): $y = \frac{7}{24}x$; thus $H = (10, \frac{35}{12})$.

The feet of the altitudes are $D = (10, 0)$ (foot from A to BC), E = foot from B to AC : projection gives $E = (\frac{9792}{625}, \frac{2856}{625})$, F = foot from C to AB : $F = (\frac{425}{169}, \frac{1020}{169})$.

Line EF meets BC (the x -axis) at X . Using the two-point form, after simplification we obtain $X = (\frac{170}{3}, 0)$.

The circumcircle Ω of ABC : its centre is the intersection of perpendicular bisectors of BC and AB . The perpendicular bisector of BC is $x = \frac{17}{2}$; the perpendicular bisector of AB (midpoint $(5, 12)$, slope of AB is $\frac{12}{5}$ so perpendicular slope $-\frac{5}{12}$) gives $y = \frac{253}{24}$ at $x = \frac{17}{2}$. Hence $O = (\frac{17}{2}, \frac{253}{24})$ and radius $R = OB = \frac{325}{24}$.

The line AX has direction vector $\vec{v} = X - A = (\frac{140}{3}, -24)$. The second intersection Q of AX with Ω is given by the chord formula $Q = A + t\vec{v}$ where $t = -\frac{2(A-O)\cdot\vec{v}}{|\vec{v}|^2}$. Compute $A - O = (\frac{3}{2}, \frac{323}{24})$ and $(A - O) \cdot \vec{v} = \frac{3}{2} \cdot \frac{140}{3} + \frac{323}{24} \cdot (-24) = 70 - 323 = -253$. $|\vec{v}|^2 = (\frac{140}{3})^2 + 24^2 = \frac{19600}{9} + 576 = \frac{24784}{9}$. Thus $t = -2(-253)/\frac{24784}{9} = \frac{506 \cdot 9}{24784} = \frac{2277}{12392}$. Therefore $Q = (10 + \frac{140}{3} \cdot \frac{2277}{12392}, 24 - 24 \cdot \frac{2277}{12392}) = (\frac{57545}{3098}, \frac{60690}{3098})$.

The nine-point circle (circle DEF) is the image of the circumcircle under the homothety with centre H and factor $\frac{1}{2}$ (it sends O to the midpoint of OH , the nine-point centre, and any point on Ω to the midpoint of the segment joining H to that point). Hence for $Q \in \Omega$ the midpoint M of HQ lies on the nine-point circle. Triangle ABC is acute (all squares of sides are less than the sum of squares of the other two), so $OH < R$ and consequently H is inside the nine-point circle. The segment HQ therefore meets the nine-point circle exactly at M . Hence $P = M$ is the midpoint of HQ .

Now compute the required ratio. Since P is the midpoint, $PQ = \frac{1}{2}HQ$ and $PE = |\frac{H+Q}{2} - E| = \frac{1}{2}|H + Q - 2E|$. Thus

$$\frac{PQ}{PE} = \frac{HQ}{2PE} = \frac{|Q - H|}{|H + Q - 2E|}.$$

Set $V = Q - H$, $W = H + Q - 2E$. Then the ratio equals $|V|/|W|$.

Compute V :

$$V = \left(\frac{57545}{3098} - 10, \frac{60690}{3098} - \frac{35}{12} \right) = \left(\frac{26565}{3098}, \frac{309925}{18588} \right).$$

Writing the first component with denominator 18588: $\frac{26565}{3098} = \frac{159390}{18588}$, so $V = \frac{1}{18588}(159390, 309925)$. Factor $159390 = 8855 \cdot 18$, $309925 = 8855 \cdot 35$ (because $8855 = 35 \cdot 253$). Hence $V = \frac{8855}{18588}(18, 35)$.

Next, $H - E = (10 - \frac{9792}{625}, \frac{35}{12} - \frac{2856}{625}) = (-\frac{3542}{625}, -\frac{12397}{7500})$. Factor $3542 = 14 \cdot 253$, $12397 = 49 \cdot 253$, and $7500 = 12 \cdot 625$, so $H - E = \frac{253}{625}(-14, -\frac{49}{12})$.

Now $W = V + 2(H - E)$. Compute squares:

$$|V|^2 = \frac{8855^2}{18588^2} \cdot (18^2 + 35^2) = \frac{8855^2}{18588^2} \cdot 1549.$$

Since $8855 = 35 \cdot 253$ and $18588 = 12 \cdot 1549$,

$$\frac{8855^2}{18588^2} = \frac{35^2 \cdot 253^2}{12^2 \cdot 1549^2} = \frac{1225 \cdot 253^2}{144 \cdot 1549^2},$$

so $|V|^2 = \frac{1225 \cdot 253^2}{144 \cdot 1549^2}$.

For $|H - E|^2$:

$$|H - E|^2 = \left(\frac{253}{625}\right)^2 \left(14^2 + \left(\frac{49}{12}\right)^2\right) = \frac{253^2}{625^2} \cdot \frac{30625}{144} = \frac{253^2 \cdot 49}{625 \cdot 144}.$$

Thus $4|H - E|^2 = \frac{196 \cdot 253^2}{625 \cdot 144}$.

For the dot product:

$$V \cdot (H - E) = \frac{8855}{18588} \cdot \frac{253}{625} (18 \cdot (-14) + 35 \cdot (-\frac{49}{12})) = \frac{8855 \cdot 253}{18588 \cdot 625} \cdot \left(-\frac{4739}{12}\right).$$

Using $8855 = 35 \cdot 253$ and $18588 = 12 \cdot 1549$, $V \cdot (H - E) = -\frac{35 \cdot 253^2 \cdot 4739}{144 \cdot 1549 \cdot 625}$, so $4V \cdot (H - E) = -\frac{140 \cdot 253^2 \cdot 4739}{144 \cdot 1549 \cdot 625}$.

Then

$$|W|^2 = |V|^2 + 4|H - E|^2 + 4V \cdot (H - E) = \frac{253^2}{144} \left(\frac{1225}{1549} + \frac{196}{625} - \frac{140 \cdot 4739}{1549 \cdot 625}\right).$$

The bracket simplifies to $\frac{405769}{1549 \cdot 625}$ because $1225 \cdot 625 + 196 \cdot 1549 - 140 \cdot 4739 = 765625 + 303604 - 663460 = 405769$. Hence

$$|W|^2 = \frac{253^2 \cdot 405769}{144 \cdot 1549 \cdot 625}.$$

Now

$$\left(\frac{|V|}{|W|}\right)^2 = \frac{\frac{1225 \cdot 253^2}{144 \cdot 1549}}{\frac{253^2 \cdot 405769}{144 \cdot 1549 \cdot 625}} = \frac{1225 \cdot 625}{405769} = \frac{765625}{405769}.$$

Since $765625 = 5^6 \cdot 7^2 = 125^2 \cdot 49$ and $405769 = 637^2$ with $637 = 7^2 \cdot 13$,

$$\frac{765625}{405769} = \left(\frac{125}{91}\right)^2.$$

Thus $\frac{|V|}{|W|} = \frac{125}{91}$, and therefore

$$\frac{PQ}{PE} = \frac{125}{91}.$$

The integers 125 and 91 are coprime, so $a + b = 125 + 91 = 216$.

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